Arbor Press



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Arbour press

What it is used for

An arbor press applies controlled manual force to press-fit, assemble, straighten, or remove suitable small components using a lever and ram. Secure the press and workpiece, select the correct tooling, and keep hands clear of pinch points to prevent crushing injuries and component ejection.

Main parts

- Rigid frame and base
- Lever or operating handle
- Ram or sliding arbor
- Ram return spring, if fitted
- Pressing attachment, punch, or adapter
- Work table, support blocks, and workpiece fixture

Before you start

1. Confirm that the press capacity and tooling are suitable for the operation and component.
2. Inspect the frame, base, lever, ram, return spring, and mounting bolts for damage or looseness.
3. Secure the press to a stable workbench or approved mounting surface.
4. Inspect the workpiece and tooling for cracks, wear, misalignment, or damage.
5. Select correctly sized supports and attachments so the force is applied along the intended axis.
6. Wear safety glasses and keep the work area clear of loose objects.

How to use

1. Position the workpiece on the press base using suitable supports or a fixture.
2. Align the component and pressing attachment so the ram force acts straight through the intended contact point.
3. Adjust the work height or support blocks with the ram fully raised and hands clear of the pinch area.
4. Keep fingers away from the ram, attachment, workpiece, and support surfaces.
5. Move the lever slowly to bring the ram into contact with the workpiece.
6. Apply force gradually and observe the component for movement, deformation, or misalignment.
7. Stop immediately if the workpiece shifts, the tooling tilts, or unusual resistance occurs.
8. Release the lever carefully and allow the ram to return as designed.
9. Confirm the component is seated or removed as intended before releasing the fixture.
10. Remove the workpiece only after the ram is fully raised and the applied force is released.

1. Inspect the finished component and clean the press and work area after use.

⚠ Safety

- Never exceed the rated capacity of the press or its attachments.
- Keep hands and body parts out of the ram's travel path and all pinch points.
- Do not use cracked tooling, unstable supports, or damaged press components.
- Do not extend the lever with a pipe or other improvised device to increase force.
- Keep the workpiece aligned and restrained to prevent it from being ejected under pressure.
- Only trained users should operate the press; do not leave it loaded or unattended.

Care & storage

- Wipe the ram, base, and frame clean after use.
- Lubricate moving parts only at the points and intervals specified by the manufacturer.
- Inspect mounting bolts, lever pivots, return spring, and ram alignment regularly.
- Replace worn or damaged attachments and keep tooling labelled and organised.
- Remove the press from service if the frame bends, the ram sticks, or the return mechanism fails.

Common mistakes to avoid

- Applying force to a misaligned workpiece or unsupported component.
- Placing fingers under the ram or near pressing surfaces.
- Exceeding the press rating by using an extended lever.
- Continuing to press when the component tilts, deforms unexpectedly, or jams.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.